Challenge 1: The Hadwiger Conjecture

Definition 1 (Graph Minor)**.** Given simple graphs X and G, we say that X is a *minor* of G if there is a function assigning to each vertex x of X a nonempty vertex set $V_x \subseteq V(G)$ such that

1. the subgraph $G[V_x]$ induced by G on V_x is connected for each x;
2. for different vertices $x, y \in V(X)$, the sets V_x and V_y are disjoint;
3. for any two adjacent vertices $x, y \in V(X)$, there are adjacent vertices $a \in V_x$ and $b \in V_y$ of G.

COMMENT FROM NATHAN: maybe we can skip the definitions of minor and chromatic number

Definition 2 (Hadwiger Number)**.** Given a graph G, the *Hadwiger number* of G is the largest integer n such that K_n is a minor of G; here K_n denotes the complete graph on n vertices. We denote the Hadwiger number by $h(G)$.

Definition 3 (Chromatic number)**.** Given a graph G and $n \in \mathbb{N}$, a *proper coloring* of G with n colours is a function $f : [n] \to V(G)$ so that adjacent vertices receive different f-values. The *chromatic number* of G is the minimum number n such that there is a proper coloring of G with n colours.

Conjecture 1.1 (Hadwiger's Conjecture [1, Conjecture 1])**.** Let $r \in \mathbb{N}$. For every finite simple graph G, we have:

$$r \leq \chi(G) \implies r \leq h(G)$$

While Conjecture 1.1 is not difficult for every small values of r, already for $r = 4$ it is equivalent to the 4-colour theorem. Robertson, Seymour and Thomas proved the above for $r = 5$ [2].

References

[1] R. Steiner. "Hadwiger's conjecture and topological bounds". In: *European Journal of Combinatorics* 122 (2024), p. 104033. ISSN: 0195-6698. DOI: https://doi.org/10.1016/j.ejc.2024.104033. URL: https://www.sciencedirect.com/science/article/pii/S0195669824001185.

[2] N. Robertson, P. Seymour, and R. Thomas. "Hadwiger's conjecture for K_6-free graphs". In: *Combinatorica* 13 (1993), pp. 279–361. DOI: 10.1007/BF01202354.